Clustering Student AI Assistant Usage Patterns Using Combined K-Means and Hierarchical Methods

1. INTRODUCTION

In recent years, AI assistants have become a common part of student life. Students use AI tools to study, write assignments, solve coding problems, and get quick answers. These AI tools help save time, improve accuracy, and support learning. Because of this, understanding how students actually use AI has become an important topic in education and data science.

Different students use AI in different ways. Some spend a long time in each session, some send many prompts, and others use AI mainly for writing or coding. These different behaviours can be grouped into patterns. Finding such patterns helps us understand learning habits and can help design better AI-based educational tools.

Problem Statement

Although many students use AI assistants, we do not clearly know how their usage patterns differ. There is no direct label saying "type of student user," so we cannot classify them using normal supervised learning. Therefore, we need unsupervised learning to discover natural groups of students based on their AI interactions.

Understanding these groups can help:

- Improve AI tools for education
- Identify heavy users vs. light users
- Understand the purpose of AI usage
- Provide better support to students

Need for the Study

Educational institutions and AI developers want to know:

- How do students interact with AI tools?
- Which types of students depend more on AI?
- Are some students using AI mainly for writing while others use it for coding?
- How satisfied are different groups of students with AI?

By clustering the students based on their behaviour, we can answer these questions clearly. This study helps improve the experience of students and guides future improvements in AI-based learning systems.

Dataset Description

The dataset contains information about student sessions with an AI assistant. Major columns include:

- **SessionID:** ID for each AI usage session
- **StudentLevel:** Undergraduate or Postgraduate
- **Discipline:** Field of study (e.g., CS, Psychology)
- **SessionLengthMin:** Total minutes spent using the AI assistant
- **TotalPrompts:** Number of questions or prompts asked
- **TaskType:** Purpose of use (Studying, Writing, Coding)
- **AI_AssistanceLevel:** How much help the student asked for
- **SatisfactionRating:** How satisfied the student was after using AI

These features help us understand how students use AI and are used for clustering.

2. OBJECTIVES:

The main objective of this study is to understand how students use an AI assistant by grouping them based on their usage patterns. Since every student uses the AI tool differently some spend more time, some ask more questions, and others use the tool for writing or coding it becomes important to find natural groups among them. To achieve this, the project uses two unsupervised learning techniques: Hierarchical Clustering and K-Means Clustering.

This study also aims to prepare the dataset properly by cleaning the data, handling missing values, converting text columns into numbers, and scaling all the features so the clustering algorithms work correctly. Another objective is to use Hierarchical Clustering to observe how the data naturally forms groups and to identify the most suitable number of clusters by studying the dendrogram. Once the number of clusters is decided, K-Means Clustering is applied to form the final groupings in a more stable and accurate way.

The project also aims to compare the results from both clustering methods to understand whether the patterns they produce are similar and reliable. Finally, the study aims to understand the meaning of each cluster by analysing what type of students fall into each group for example, whether they are long-session users, quick-answer seekers, writing-focused users, or coding-heavy users. These insights will help educators and AI developers understand how students interact with AI tools and how these tools can be improved in the future.

3. METHODOLOGY:

This chapter explains how the whole project was done, step by step. The goal of the study is to group students based on how they use an AI assistant. To do this, two clustering methods were used: **Hierarchical Clustering** and **K-Means Clustering**. Both methods help find patterns in the data without using any labels.

a. Data Cleaning

The first step was to clean the dataset. The data was checked for missing values, duplicate entries, and incorrect information. If any values were missing, they were filled with the average or most common value. Duplicate rows were removed to make sure the dataset was correct and usable.

b. Converting Text to Numbers

Some columns in the dataset, like *Task Type, Discipline,* and *Student Level*, contained words. Since clustering works only with numbers, these columns were converted into numeric form using simple encoding techniques. This step makes the data ready for machine learning.

c. Selecting Important Features

Not all columns in the dataset were needed for clustering.
Only the features that describe student behaviour were selected, such as:

- Session length (time spent using AI)
- Total prompts (how many questions they asked)
- AI assistance level (how much help they needed)
- Task type (studying, writing, coding)
- Satisfaction rating

These features help show how students actually use the AI assistant.

d. Scaling the Data

All the selected features were then scaled so they have similar ranges. This is important because clustering uses distance, and one feature should not dominate the others. Scaling makes all features fair and equal.

e. Hierarchical Clustering

Next, Hierarchical Clustering was used to understand how the data naturally groups. A dendrogram (tree-like diagram) was drawn to see how students merge into groups. By looking at the height of the branches in the dendrogram, a suitable number of clusters was chosen. This helps decide how many groups exist in the data.

f. Choosing the Number of Clusters

The dendrogram from hierarchical clustering helped decide how many clusters (groups) should be formed. The largest gap in the dendrogram shows the best place to split the data into meaningful groups.

g. K-Means Clustering

After deciding the number of clusters, K-Means Clustering was used to form clearer, more stable clusters.

- K-Means grouped the students into clusters based on how similar their usage patterns were.
- Each student session was assigned to one of the groups.
- The clustering quality was checked using scores like the silhouette score to make sure the groups were separated well.

h. Comparing Both Methods

Both clustering methods were compared to see if they produced similar patterns. Hierarchical clustering helped understand the structure, and K-Means gave the final, well-separated groups. Using both methods together made the results more reliable.

i. Cluster Profiling

The final step was to understand what each cluster means. For each cluster, the average session length, number of prompts, task types, satisfaction ratings, and AI assistance levels were studied.
This helps describe each group, such as:

- Students who use AI for long periods
- Students who ask many prompts
- Students who mostly write or code
- Students who depend heavily on AI

This profiling gives clear insights into how students behave while using the AI assistant.

4. RESULTS AND ANALYSIS:

This chapter presents the results obtained from the hierarchical and K-Means clustering methods that were applied to group students based on their AI assistant usage patterns. The analysis includes how the number of clusters was selected, how the clusters were formed, and what differences were observed between them.

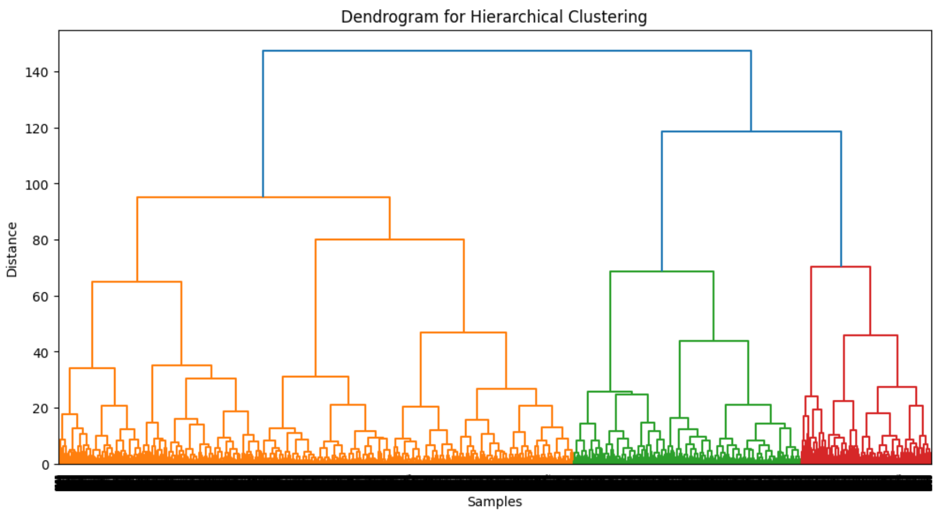
4.1 Dendrogram Analysis (Hierarchical Clustering)

To understand how many natural groups exist in the dataset, **Hierarchical Clustering** was first applied using the **Ward linkage method**. Ward's method merges clusters by minimizing the increase in total variance, which produces compact and meaningful groups suitable for numerical, scaled data.

The **dendrogram** generated from this method provides a tree-like structure showing how data points merge step by step. Each vertical line represents a merge, and the height of the merge indicates how different the clusters are from one another.

In the dendrogram for this dataset, a **clear separation** can be seen at the upper level of the tree, where the dataset divides into **three major branches**. The large vertical distance before the final merges indicate that combining beyond three clusters would merge groups that are naturally very different.

Therefore, the dendrogram strongly suggests that the **optimal number of clusters is 3**. This forms the basis for selecting the value of *k* in the next stage using K-Means clustering.



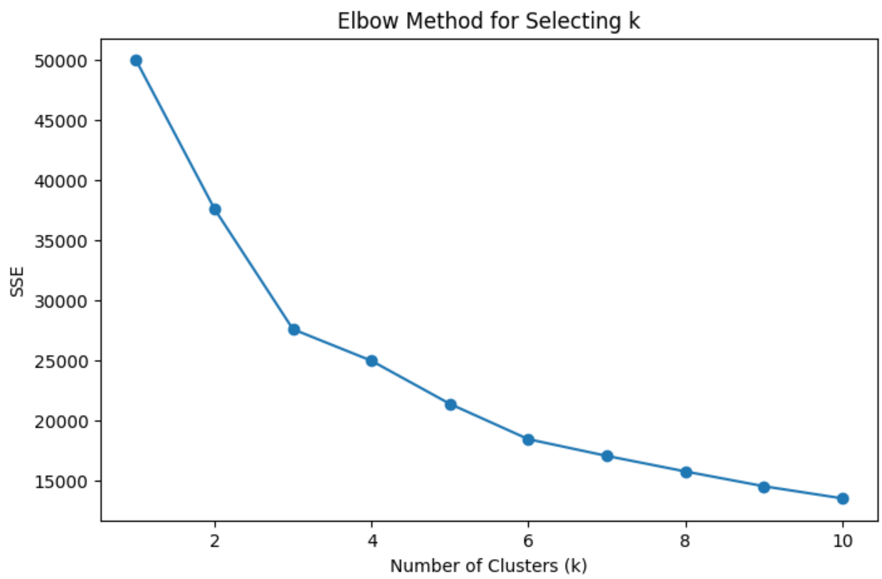
4.2 Elbow Method Analysis (K-Means Clustering)

After identifying the natural structure of the dataset through the dendrogram, the next step was to confirm the number of clusters using the **Elbow Method**. This method helps identify the value of *k* (number of clusters) where adding more clusters no longer significantly improves the clustering quality.

In the Elbow Method, the **Sum of Squared Errors (SSE)** is calculated for different values of *k*. SSE represents how far data points are from their assigned cluster centres. A lower SSE is better, but the rate of improvement decreases after a certain point.

In the elbow plot generated for this dataset, SSE drops sharply from **k = 1 to k = 3**, and after **k = 3**, the curve begins to flatten. This bending point in the graph is known as the "elbow." The flattening after k = 3 indicates that adding more clusters does not significantly reduce the SSE.

Therefore, the elbow method confirms that **k = 3** is the most appropriate and efficient number of clusters to use for K-Means clustering.



4.3 Silhouette Score Analysis

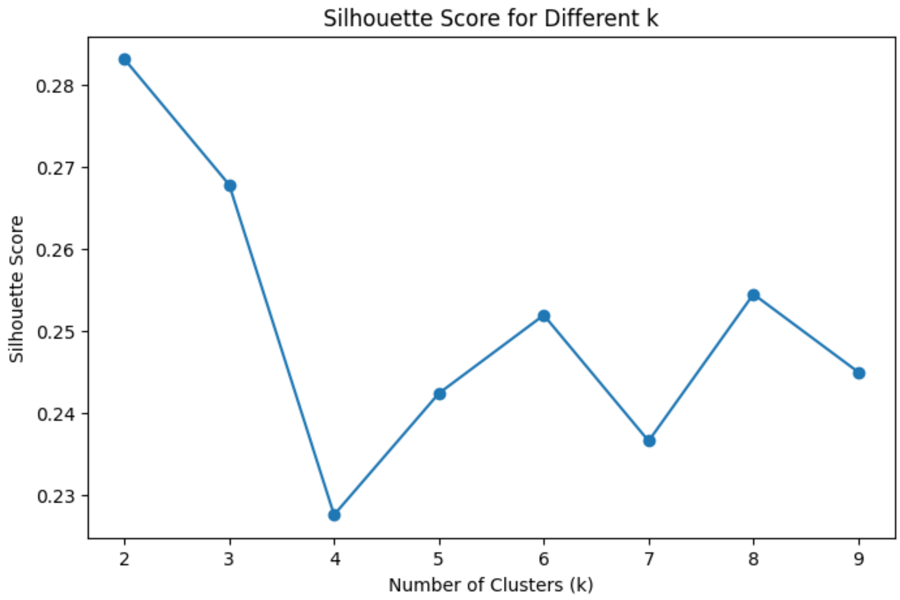
The **Silhouette Score** is another important method used to evaluate the quality of clusters. It measures how well each data point fits within its assigned cluster compared to other clusters. The score ranges from **-1 to +1**, where:

- **+1** → Data points are well-matched to their cluster
- **0** → Data points lie between two clusters
- **Negative values** → Data points may be assigned to the wrong cluster

For this dataset, silhouette scores were calculated for different values of k (from 2 to 9). The plot shows that the **highest score occurs at k = 2**, but the score for **k = 3** is only slightly lower and still relatively high.

Even though k = 2 gives the highest score, using two clusters would oversimplify the dataset and merge distinct behavioural groups. Both the dendrogram and elbow method strongly pointed to **three natural clusters**, and silhouette score for k = 3 is also acceptable.

Therefore, considering all evaluation methods together, **k = 3** is confirmed as the most meaningful and balanced choice for clustering this dataset.



4.4 Cluster Size Analysis

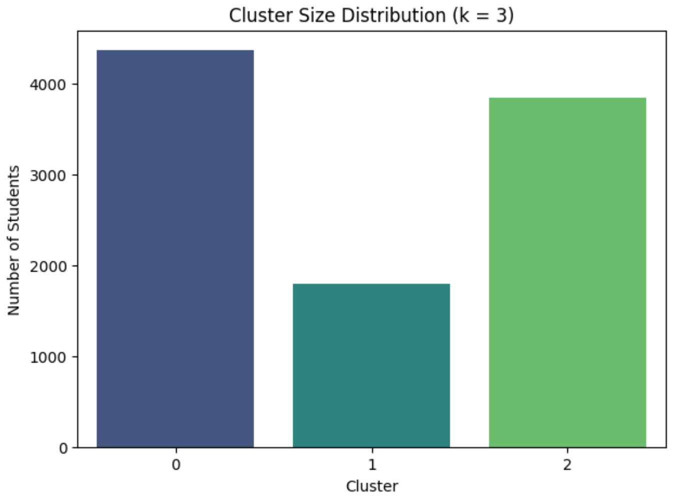
After applying K-Means clustering with **k = 3**, the dataset was successfully divided into three distinct student groups. Understanding the size of each cluster is important because it shows how many students fall under each usage pattern.

The cluster size distribution shows the following:

- **Cluster 0:** 4364 students
- **Cluster 1:** 1790 students
- **Cluster 2:** 3846 students

Cluster 0 is the largest group, followed closely by Cluster 2, while Cluster 1 is the smallest group. This indicates that most students fall into two major usage patterns, while a smaller but significant group shows different behaviour.

The cluster size bar chart clearly shows that each cluster contains a good number of students, meaning the clustering is balanced and meaningful. None of the clusters are too small or too large, which confirms that **k = 3** is an appropriate choice for this dataset.

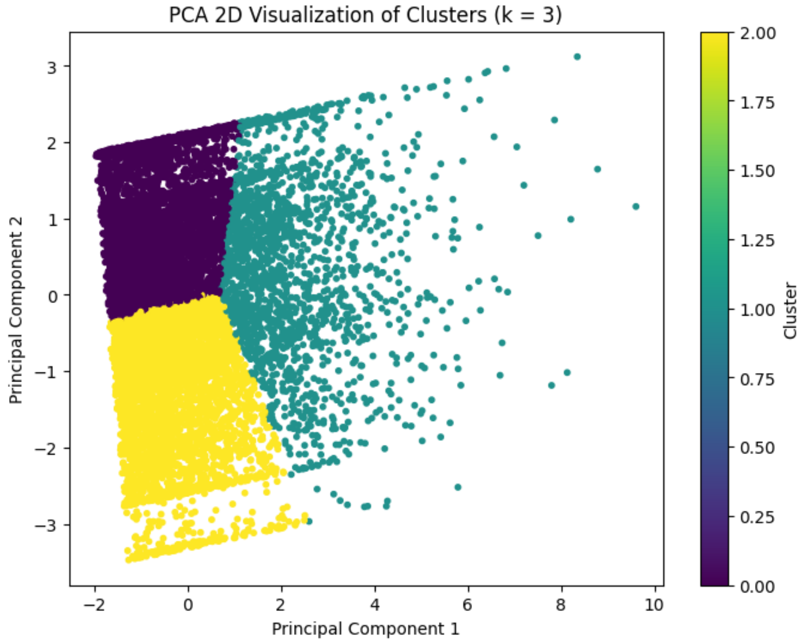


4.5 PCA Visualization of Clusters

To visually confirm how well the clusters are separated, **Principal Component Analysis (PCA)** was applied to reduce the high-dimensional dataset into two principal components. These two components capture most of the variation in the data and allow the clusters to be displayed on a simple 2D scatter plot.

The PCA scatter plot for the three clusters shows **clear separation** between the groups. The clusters appear as distinct coloured regions, indicating that the K-Means algorithm has effectively grouped students based on their AI usage behaviour. The boundaries between clusters are well-defined, and there is minimal overlap, which confirms that the chosen features and the number of clusters (k = 3) are suitable for this dataset.

This visualization provides strong evidence that the clustering performed in the earlier steps is meaningful and that each cluster represents a different type of student usage pattern.



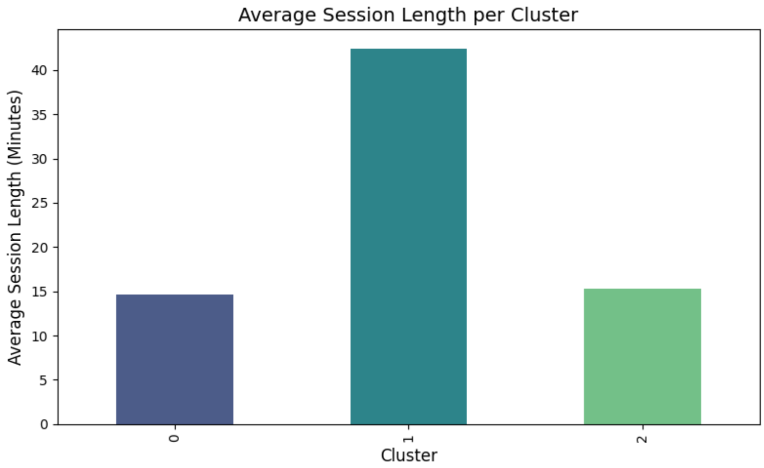
4.6 Session Length Analysis

Session length tells us how long students use the AI assistant in a single session. To compare the average session duration across the three clusters, a bar chart was created.

The results show a clear difference:

- **Cluster 1 has the longest average session length**, with students spending around 42 minutes in a session. This means they interact with the AI assistant for a longer time and show deeper engagement.
- **Cluster 0 and Cluster 2 have much shorter average session lengths**, around 14 to 15 minutes. These students use the AI assistant for quick and short interactions.

The bar chart makes it easy to see that Cluster 1 is the high-engagement group, while Clusters 0 and 2 are short-duration users. This difference helps us understand how intensively each cluster uses the AI tool.



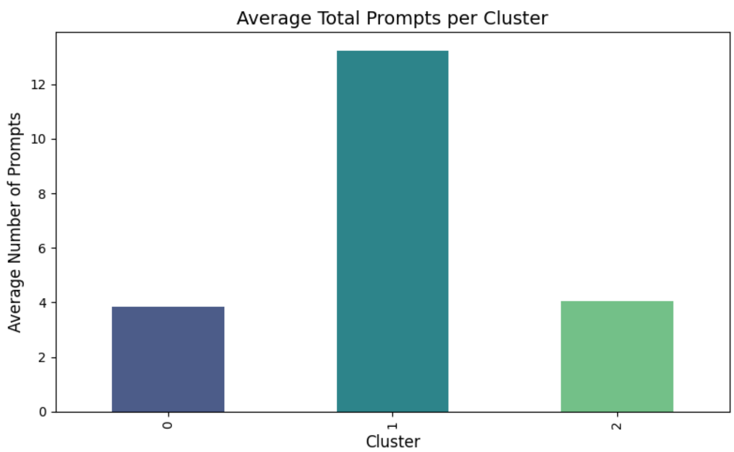
4.7 Total Prompts Analysis

Total prompts represent how many questions or messages a student sends to the AI assistant in one session. This helps us understand how actively each group of students interacts with the AI tool. To compare this across the three clusters, we calculated the average number of prompts in each group and plotted it using a bar chart.

The results are clear:

- **Cluster 1 has the highest average number of prompts** (about 13). Students in this group interact with the AI assistant the most and ask many follow-up questions.
- **Cluster 0 and Cluster 2 have much lower average prompts** (around 3–4 each). These groups use the AI briefly and send fewer messages.

This shows that Cluster 1 is the highly interactive user group, while Clusters 0 and 2 represent short-interaction or low-engagement users.



4.8 Task Type Distribution Across Clusters

Students use the AI assistant for different types of tasks. In the dataset, each task is represented by a number (0 to 5). To understand how each cluster uses the AI for different purposes, a bar chart was created showing the count of each task type in every cluster.

From the chart:

Cluster 0

- Uses a **wide range of tasks**.
- Task Types **1, 2, 4, and 5** are especially high.
- This means Cluster 0 students use the AI assistant for **many different activities**, such as writing, studying, summarizing, etc.

Cluster 1

- Has fewer overall counts because the cluster itself is smaller.
- Still uses a mix of tasks, but less variety than Cluster 0 and 2.
- Focused mainly on Task Types **1, 2, and 5**.
- These students interact deeply (many prompts), even if task variety is smaller.

Cluster 2

- Very similar variety to Cluster 0.
- High counts in Task Types **1, 2, 4, and 5**.
- Shows that this group also uses the AI assistant for many purposes.

Simple Meaning

- **Clusters 0 and 2** → Students use the AI for **many different kinds of tasks**.
- **Cluster 1** → Students use the AI for **fewer types of tasks**, but they use it **more deeply** (more prompts, longer sessions).

This tells us the type of tasks students do depends on the cluster, and different groups use the AI for different reasons.

Task Type 0	Brainstorming
Task Type 1	Coding
Task Type 2	Explanation
Task Type 3	Studying
Task Type 4	Summarization
Task Type 5	Writing

4.9 AI Assistance Level Analysis

AI Assistance Level tells us how much help students requested from the AI assistant during their session. The values range from low assistance to high assistance. To compare how much support each cluster needed, the average AI assistance level was plotted.

From the results:

Cluster 0 – Highest Assistance

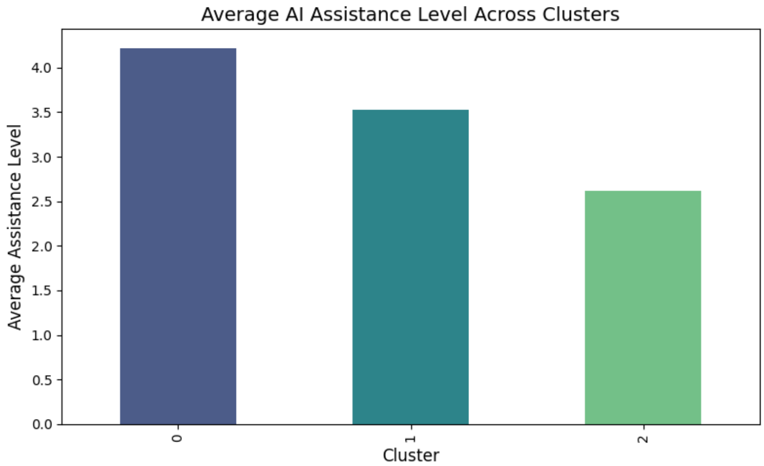
Students in Cluster 0 needed the **most help from the AI assistant**. Their average assistance level is the highest among all three groups. This means they rely on the AI for explanations, solving tasks, writing help, and study support.

Cluster 1 – Medium Assistance

Cluster 1 students have a **moderate assistance level**. Even though they use the AI for longer sessions and ask many prompts, they do not ask for maximum help. They explore tasks more independently.

Cluster 2 – Lowest Assistance

Cluster 2 shows the **lowest assistance level**. These students interact less and depend less on the AI for difficult tasks, guidance, or corrections.



4.10 Satisfaction Rating Analysis

Satisfaction Rating shows how happy or satisfied students are with the AI assistant after using it. Higher ratings mean the students were more satisfied with the help they received.

To compare satisfaction across the three clusters, the average satisfaction score was calculated and shown using a bar chart.

The results are easy to understand:

Cluster 0 – Highest Satisfaction

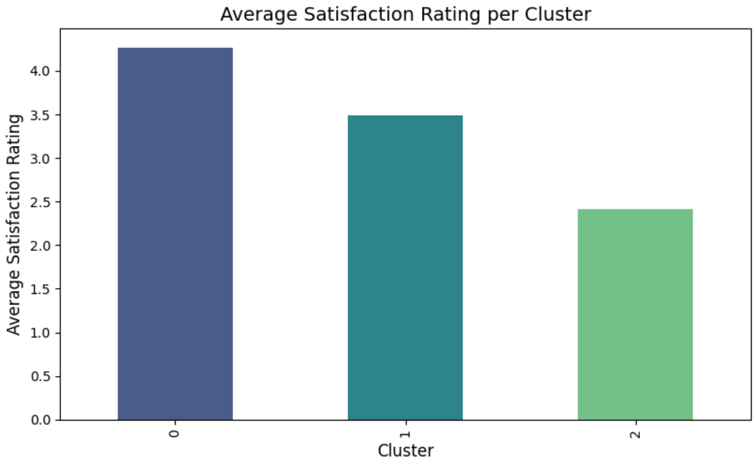
Cluster 0 has the **highest satisfaction rating** among all groups. These students also have high assistance levels, which suggests they get helpful responses that meet their needs.

Cluster 1 – Medium Satisfaction

Cluster 1 has a **moderate satisfaction rating**. They interact more and spend more time with the AI, but their satisfaction is not the highest. This may be because they work on more complex tasks.

Cluster 2 – Lowest Satisfaction

Cluster 2 has the **lowest satisfaction rating**. They spend less time and ask fewer prompts, and their low satisfaction may suggest less benefit from the AI or less interest in using it.



4.11 Summary Table Interpretation

To understand the overall behaviour of each cluster more clearly, the main features Session Length, Total Prompts, AI Assistance Level, Task Type, and Satisfaction Rating were summarized into a single table.

This summary helps us compare all three clusters at once and see the differences in their usage patterns.

From the summary values:

Cluster 0

- Short session time
- Low number of prompts
- Highest AI assistance level
- Highest satisfaction rating

This cluster relies heavily on the AI assistant and is very satisfied with the help.

Cluster 1

- Longest session length
- Highest number of prompts
- Medium assistance level
- Moderate satisfaction

These students interact deeply with the AI for a long time but are not the most satisfied.

Cluster 2

- Short session time (similar to Cluster 0)
- Low prompts
- Lowest assistance level
- Lowest satisfaction rating

These students use the AI lightly and gain less benefit from it.
This table gives a clear comparison of all three student groups and their usage patterns.

Cluster	SessionLengthMin	TotalPrompts	AI_AssistanceLevel	TaskType	SatisfactionRating
0	14.59	3.84	4.22	3.12	4.27
1	42.44	13.24	3.52	2.97	3.49
2	15.3	4.06	2.62	3.13	2.42

4.12 Cluster Interpretation (Student Personas)

The three clusters represent **three types of student behaviours**:

1. **Cluster 0: Support-Seekers** → Need help, trust AI, satisfied.
2. **Cluster 1: Deep Explorers** → Spend long time, ask many prompts, moderately satisfied.
3. **Cluster 2: Light Users** → Use AI less, depend less, least satisfied.

These personas help understand how students interact with AI differently and how support systems can be improved for each group.

5. CONCLUSION:

This project aimed to group students based on how they use the AI assistant by applying two clustering techniques: **Hierarchical Clustering** and **K-Means Clustering**. Using the dendrogram, elbow method, and silhouette score, the project identified that the dataset naturally forms three meaningful clusters.

Each cluster showed a different pattern of AI usage:

- **Cluster 0 students** depend highly on the AI assistant and are highly satisfied with its support.
- **Cluster 1 students** spend the most time interacting with the AI and ask many prompts, showing deep engagement.
- **Cluster 2 students** use the AI lightly, depend less on it, and have the lowest satisfaction.

These findings clearly show that different types of students interact with AI in different ways. Some use it as a strong support tool, some use it for long and continuous study, while others use it only occasionally.

The combination of **Hierarchical and K-Means Clustering** made the results more reliable. Hierarchical clustering helped correctly identify the number of clusters, and K-Means provided clean, well-separated final groupings. Overall, the project successfully achieved its objective of understanding student behaviour patterns based on AI assistant usage.

Final Summary

The main problem addressed in this study was the lack of clarity regarding how students use AI assistants in different ways. Although AI tools are widely used in education, the dataset did not contain any predefined labels such as “heavy user,” “light user,” or “study-focused user.” Because of this, supervised learning methods could not be used. To overcome this challenge, the study applied unsupervised learning techniques to automatically identify natural groups of students based on their AI usage patterns.

Using hierarchical clustering and K-Means clustering, the study successfully identified **three distinct clusters** of student users. These clusters represent meaningful patterns of behaviour:

- **Cluster 0:** Students who depend heavily on AI support and show high satisfaction levels.
- **Cluster 1:** Students who spend the most time interacting with the AI, ask many prompts, and show deep engagement.
- **Cluster 2:** Students who interact with the AI minimally and report low satisfaction.

These results clearly answer the problem statement by showing that students do not use AI in the same way. Instead, they fall into three natural categories based on their interaction levels, task types, and satisfaction ratings. The clustering approach allowed the study to uncover these hidden patterns without the need for labelled data.

Overall, the project achieved its objective by providing a clear understanding of the different types of AI users among students. This knowledge can help educators and developers improve AI tools, offer personalized learning support, and better meet the needs of different student groups. The study demonstrates that unsupervised learning is a powerful method for analysing user behaviour when labels are not available.